

Success of restoration strategies in preventing extirpation of 2 critically endangered coral species

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Abstract

An unprecedented marine heatwave in 2023 caused widespread coral bleaching and mortality throughout the Caribbean. In the Florida Keys (USA), 2 foundation species, elkhorn coral (*Acropora palmata*) and staghorn coral (*Acropora cervicornis*), were severely affected. These species have been the primary focus of reef restoration in Florida for decades. Substantial losses of these species occurred in outplanted populations, in ocean-based nurseries, and among remnant wild colonies, leading to uncertainty over their future in the Florida Keys, given recent observed trends in climate conditions. However, the past 2 decades of restoration activity created a community of experts, a network of ocean-based and land-based coral-rearing infrastructure, and 2 independent land-based coral gene banks that prevented regional extirpation and preserved much of the genetic richness of these critically endangered coral species. Without the past decades of effort and the emergency response associated with the 2023 bleaching event, Florida acroporids would largely have been lost. This outcome afforded by the restoration network in Florida demonstrates the value of proactively establishing resources prior to major disturbances. We identified several critical strategies that, in the context of existing climate change, are preventing the extirpation of coral species in Florida, including extending collaborative restoration efforts to solidify a network of trained experts; establishing trust-focused relationships among management agencies and restoration groups; testing direct interventions to reduce light and temperature stress early during thermal anomalies; developing redundant ocean-based and land-based nurseries; and establishing living coral gene banks prior to major threats to prevent the extirpation of coral species.

KEYWORDS

Acropora cervicornis, *Acropora palmata*, conservation planning, coral reef, coral restoration, threatened species

INTRODUCTION

Coral cover has declined significantly throughout Florida's Coral Reef over the last 50 years (Porter & Meier, 1992; Ruzicka et al., 2013). Major losses are largely attributable to regional coral bleaching events (Manzello et al., 2007; Somerfield et al., 2008; Wagner et al., 2010; Williams et al., 2017) and infectious coral diseases (Muller et al., 2020; Porter et al., 2001; Precht et al., 2016). Additionally, reduced water quality (Wagner et al., 2010), overfishing (Zaneveld et al., 2016), ocean acidification (Muehllehner et al., 2016), land use (Carlson et al., 2019), and other threats (Nalley et al., 2021) likely have had indirect effects on corals by inhibiting reef recovery or increasing reef-erosion rates. Current estimates suggest that coral cover is ~2% throughout Florida's Coral Reef (Jones et al., 2022), equivalent to over 90% relative reduction in coral cover. This reduction has likely led to the loss of several critical ecosystem functions (Swaminathan et al., 2024; Toth et al., 2023). To date, there have been little to no signs of recovery, even with the establishment of many well-connected marine protected areas throughout the region (Toth et al., 2014). Although some locally managed threats have been reduced, such as centralized sewage treatment (Kelly et al., 2021), additional efforts to help reef ecosystems and species persist until global anthropogenic impacts are mitigated now include the propagation and restoration of coral populations (Kleypas et al., 2021). In Florida, local nonprofits, universities, and government-led organizations—including the National

Oceanic and Atmospheric Administration (NOAA), the Florida Fish and Wildlife Conservation Commission (FWC), and the Florida Department of Environmental Protection (FDEP)—have implemented small-scale pilot projects to large-scale regional efforts (www.fisheries.noaa.gov/iconic-reefs). These approaches are now being adopted around the globe (Cook et al., 2022; McLeod et al., 2022).

Staghorn (*Acropora cervicornis*) and elkhorn (*Acropora palmata*; Lamarck, 1816) have been the focus of restoration in Florida and the Caribbean for decades (Boström-Einarsson et al., 2020; Lirman & Schopmeyer, 2016). This effort emerged from concerns regarding the Caribbean acroporids endangered species status and legal designations, which prompted investment in species-focused restoration actions beginning in the early 2000s, before coral restoration was considered an appropriate management action (Precht et al., 2005). Both acroporids are key reef-building species that form extensive thickets and historically characterized entire Caribbean reef ecozones. NOAA designated both as candidate species under the *United States Endangered Species Act* (ESA) in 1999 and listed them as threatened under the ESA in 2006. Critically endangered status was conferred for both species in 2008 by the International Union for Conservation of Nature (Bruckner, 2002; Carpenter et al., 2008; Hogarth, 2006). These official designations and the inclusion of active species restoration in the official U.S. recovery plan (NOAA, 2015) encouraged substantial public and private investment in restoration actions. Coral propagation and restoration of *Acropora*

spp. have since been acknowledged as standard management practice.

The summer of 2023 brought a record-breaking marine heatwave that exceeded heat stress accumulation by 3 times any previous record in the Caribbean. Water temperatures in Florida Coral Reef have been increasing rapidly (Manzello, 2015). The previous major bleaching event yielded mass mortality of Florida *Acropora* spp. in 2014–2015 (Williams et al., 2017). Satellite-derived estimates of maximum heat stress accumulation along the Florida Keys were approximately 7–8 degree heating weeks (dhw) (https://coralreefwatch.noaa.gov/data/5km/v3.1_op/image/daily/dhw/png/2014/09/ct5km_dhw_v3.1_florida_20140901.png). The similarly derived estimates for 2023 were in the range of 18–20 dhw along the main Florida Keys and 10–13 dhw along southeast Florida (https://coralreefwatch.noaa.gov/data/5km/v3.1_op/image/composite/monthly/dhw-max/png/2023/ct5km_dhw-max_v3.1_sflorida_202309.png). The early onset and duration of severe heat stress during the summer of 2023 were also unprecedented, with dhw accumulation beginning in June, persisting through October, and substantial bleaching and mortality of acroporids underway by mid-July (Williams et al., 2024).

The result of the 2023 bleaching event resulted in the near extirpation of acroporid corals in much of Florida's Coral Reef regardless of whether they were outplanted to reef sites (all authors, personal observation) or the few remaining wild colonies (Williams et al., 2024). Without the last 2 decades of restoration efforts, the development of a trained workforce of experts, the communication between local management and restoration groups, and a network of land-based infrastructure including redundant living coral biobanks, these 2 foundation species would have been largely extirpated from Florida. Indeed, there is no better example of the undefined and indirect value of restoration efforts than the outcome of the cooperation of Florida's restoration community during the summer 2023 marine heatwave. The activities described here represent the primary actions used to preserve these critical species in Florida and provide a reflection on lessons learned and recommendations for other regions to consider.

DECADES OF CORAL RESTORATION CREATED INHERENT REDUNDANCY

A regional network of ocean-based nurseries

Over the past 2 decades, ocean-based coral nurseries have emerged as pivotal tools for producing high amounts of biomass. These structures were created as a cost-effective method for growing and propagating fast-growing branching corals, such as *A. cervicornis* (Figure 1a), and then later used with *A. palmata* to foster high coral survival and rapid growth (Johnson et al., 2011; Lirman et al., 2010; Nedimyer et al., 2011). There is now a network of >20 ocean-based nurseries across southeast Florida from Broward County to Dry Tortugas National Park that are managed by a diverse consortium of

stakeholders, including nonprofit organizations, academic institutions, and state and federal agencies. This collaborative effort allowed for the production and outplanting of hundreds of thousands of coral fragments annually of over 23 species under the management of coral restoration specialists (Figure 2).

The significance of these ocean-based nurseries extends beyond their capacity for asexual propagation. The crucial role of ocean-based nurseries as genetic repositories was evidenced as early as the 2010 cold-water anomaly that caused significant mortality throughout Florida's Coral Reef (Colella et al., 2012; Lirman et al., 2011). During this event, genets (i.e., unique genetic individuals) of *A. cervicornis* died on their reefs of origin but survived in nurseries. Similar situations have subsequently occurred in relation to hurricanes, bleaching events, and disease outbreaks. These surviving native genets have served as a source to repopulate depleted reefs (Schopmeyer et al., 2012).

Land-based infrastructure for support and propagation

In addition to ocean-based coral nurseries, several land-based coral nurseries were established prior to 2023 as a complement to the traditional ocean-based nurseries (Figure 1b). Land-based infrastructure provides unique advantages to restoration practitioners because water quality and environmental conditions can be controlled (including protection from disturbances such as disease and thermal anomalies); coral care is more readily accomplished because there are no field access limitations; and the accessibility allows for more direct and inclusive hands-on education and volunteer opportunities. The versatility of land-based holding was also adopted for other purposes, including assisted sexual reproduction of corals (Craggs et al., 2020) as a means for continual stocking of coral nurseries. Land-based spawning systems mimic environmental cues to encourage the gametogenesis process in corals, yielding eventual release of gametes (spawning species) or competent larvae (brooding species). However, there are limitations associated with this type of infrastructure including expense, space, expertise, reliable power, and risks associated with disease that are found in high-density aquaculture practices (Sheridan et al., 2013). Regardless, having a diverse approach to restoration has proven increasingly useful for propagating corals for restoration. Presently, there are almost a dozen locations certified as holding facilities associated with restoration for Florida-sourced corals (Figure 2).

Coral swaps

As wild populations of both acroporids in Florida declined over the last several decades, more targeted genetic management planning began. Starting in 2015, coral nursery practitioners were considering the importance of safeguarding genetic diversity of *A. cervicornis* by replicating genets (eventually identified using tools created by Kitchen et al. [2020]) in more than one ocean-based nursery location. Over the years since initiation, there has been an increased focus on this effort, particularly

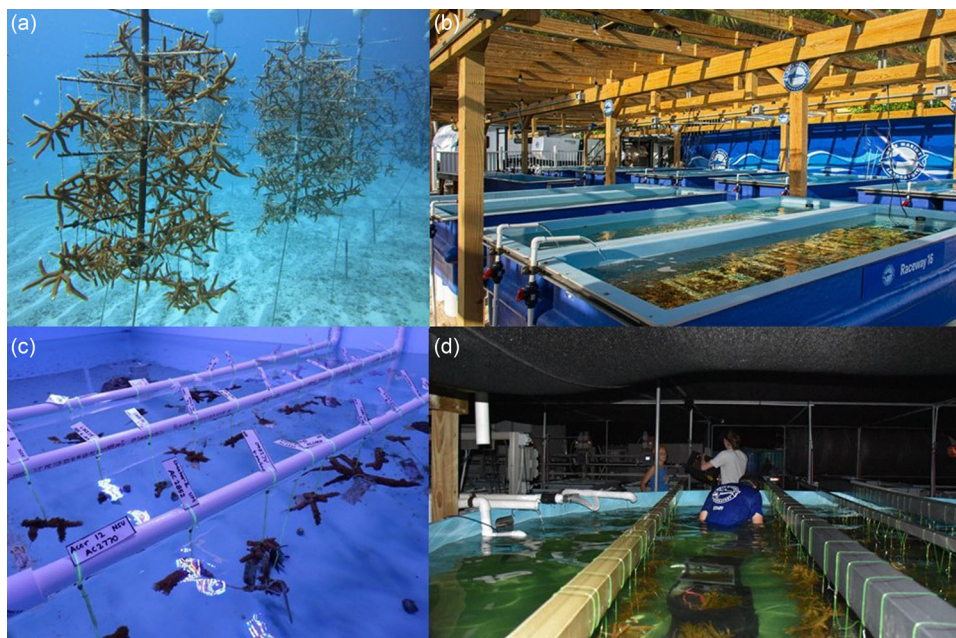


FIGURE 1 (a) Underwater coral nursery propagating acroporid corals on the commonly used tree structure, (b) land-based coral nursery growing fragments of *Acropora* spp., (c) corals received for live coral biobanking, and (d) evacuated acroporid corals being cared for within a land-based nursery. Photos by Mote Marine Laboratory.

for these ESA-listed species. Thousands of coral colonies from hundreds of distinct genets were exchanged between ocean-based nurseries (Caitlin Lusic, personal communication). This was conducted systematically such that ramets (i.e., physically isolated coral fragments with the same genetic composition created through asexual fragmentation) of most *A. cervicornis* genets were held in at least 3 different ocean-based nursery locations, reducing the risk that any given disturbance (e.g., hurricanes, disease outbreaks, pollution events, thermal anomalies, etc.) would result in total loss of a genet.

Outplant populations

Over the past decade, hundreds of thousands of acroporid fragments were outplanted to Florida's Coral Reef from Broward County to the Dry Tortugas by numerous restoration groups. Although many corals were lost within years of outplanting (Boisvert et al., 2024; Ware et al., 2020; van Woesik et al., 2021), many survived and were a repository of biomass for additional nursery collections (Schopmeyer et al., 2012). Outplant sites can also act as hubs for spawning activity (see "IMPORTANCE OF CORAL RESTORATION TO INCREASE GENETIC REDUNDANCY AND SEXUAL REPRODUCTION"), both natural and facilitated. Spawning hubs can support assisted sexual reproduction for species that are difficult to spawn in land-based facilities (e.g., *A. palmata*) and can be used to colocate broodstock either for natural fertilization, for direct collection of spawn, or as sources of branches that could be integrated into land-based spawning efforts. Because little to no natural recruitment is occurring for acroporid species in Florida (Harper et al., 2023; Williams et al., 2008), conducting assisted sexual

reproduction from outplanted corals provided a vital source of genetic recombination needed to facilitate the opportunity for natural selection and for further research and restoration efforts.

IMPORTANCE OF CORAL RESTORATION TO INCREASE GENETIC REDUNDANCY AND SEXUAL REPRODUCTION

The general decline of *A. palmata* populations in Florida over several decades rendered them largely incapable of natural larval production due to low abundance and genet density (Williams et al., 2020), prompting extensive propagation and outplanting efforts in high densities beginning around 2015. For example, the small patch reef at North Dry Rocks had 27 live *A. palmata* colonies in 2014, down from 74 in 2006 (Miller et al., 2016). This reef received *A. palmata* outplants in 2015 (120 fragments, 4 genets) and 2018 (1050 fragments, 50 genets). From 2020 to 2022, a subset of these colonies spawned annually, producing gametes that were successfully collected and crossed from over 20 genets originating from 12 different reefs throughout the Florida Keys (~200 km) ultimately producing viable offspring from this restored population (Williams et al., 2023). Similarly, multiple genets of *A. palmata* outplanted to Elbow Reef in 2016 (288 colonies, 8 genets) spawned synchronously with wild colonies at this site as early as 2021, again producing viable larvae from gamete collections.

Coral restoration in Florida has achieved ecologically meaningful success at reef-site spatial scales in several instances by producing genetically diverse sites that are reproductively active (Figure 3). Until the marine heatwave of 2023, *A. palmata* could be outplanted in Florida to create genetically diverse thickets

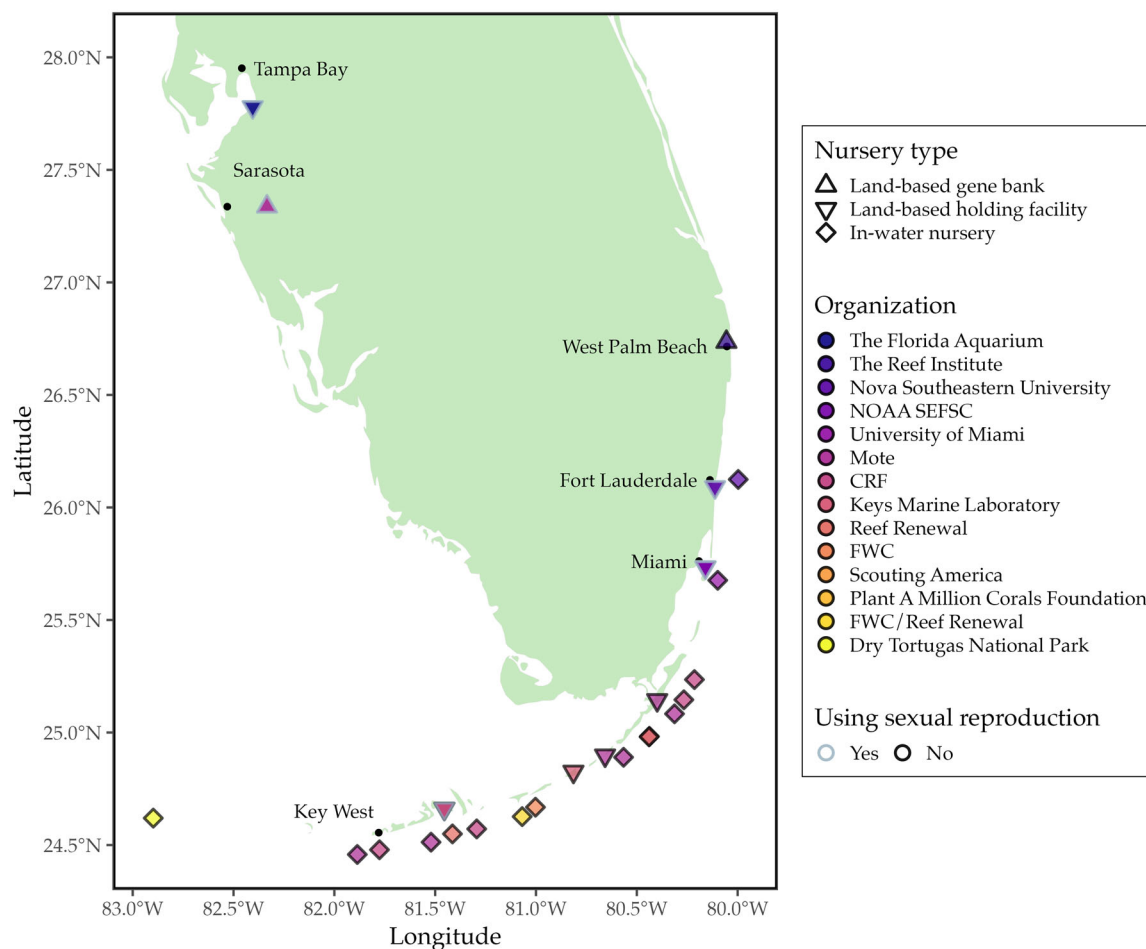


FIGURE 2 Locations of coral nurseries throughout Florida at the time of the emergency coral response and rescue event in 2023 (organizations in the key are listed north to south; NOAA SEFSC, National Oceanic and Atmospheric Administration Southeast Fisheries Science Center; Mote, Mote Marine Laboratory; CRF, Coral Restoration Foundation; FWC, Florida Fish and Wildlife Conservation Commission).

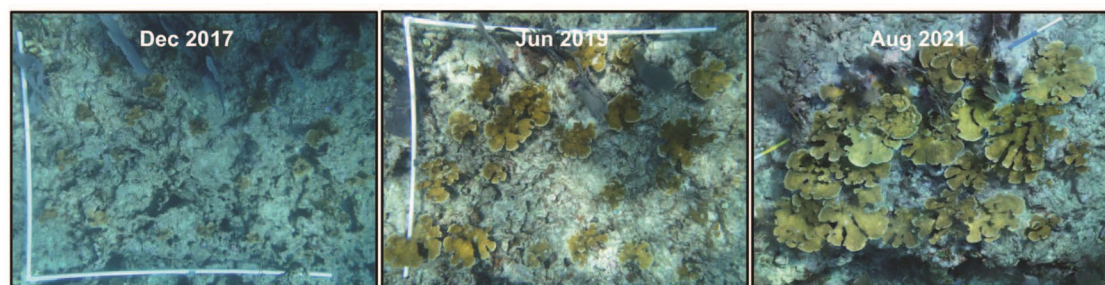


FIGURE 3 Time-series images of outplanted *Acropora palmata* (original outplanting in 2016) showing, from left to right, the growth from fragments to establishment of reproductive-sized individuals that spawned repeatedly from 2020 to 2023 (photo by NOAA Southeast Fisheries Science Center).

that successfully spawned and led to the potential crossing of up to 45 genets. Many of these gametes would never have been fertilized without intervention due to distances of tens to hundreds of kilometers between wild colonies on native reefs. Even with the unprecedented early onset of thermal stress in 2023, gametes were still collected from restored colonies at North Dry Rocks in August 2023 (M.W.M., personal observation). Ulti-

mately, all of the ~125 colonies, consisting of 45 unique genets, that were alive in June 2023 survived 5–8 years after outplanting, succumbed to thermal stress, and died by the end of the summer (M.L., personal observation). Due to their previous success, the reestablishment of these spawning hubs has been recognized as a near-term priority for the restoration community. Because all the existing genets are redundantly represented

in gene banks, nurseries, or both, if the coral designated for a new spawning hub succumbs to another disturbance event, little will be lost, and in the meantime, thousands of coral recruits could be created.

COORDINATED RESPONSE TO THE 2023 HEAT EVENT

Although disturbance-response action plans are often incorporated into reef restoration frameworks, significant resources (including time) are needed to execute such plans (Carrick et al., 2019). The magnitude and potential impacts of the 2023 mass bleaching event, however, crossed the threshold beyond which inaction would have invariably resulted in devastating population-level losses and motivated the largest coral evacuation ever attempted. Beginning in late July 2023, all reef restoration stakeholders (i.e., academia, nongovernmental organizations, local aquaria, government agencies, and the public) conducted an emergency coral response and rescue effort throughout Florida to mitigate losses of coral genetic diversity and biomass during the extreme temperature anomaly. Resources were rapidly mobilized to evacuate corals from multiple ocean-based coral nurseries throughout Florida in a 2-stage approach. First, NOAA coordinated all groups to collect living samples of as many wild-collected (i.e., founder) genets in culture as possible across both acroporid species for the development of living coral biobanking to safeguard genotypic richness (Figure 1c). Second, a larger effort to evacuate and preserve biomass from ocean-based nurseries was initiated in hopes that restoration operations could resume quickly following the thermal stress event (Figure 1d).

Coral biobanking

Since ~2018, the need to create land-based living coral gene banks was a priority for the restoration community to secure a safe haven for species preservation. Efforts created solely for this purpose currently exist in Florida (Mote Marine Laboratory's International Coral Gene Bank), Monaco (World Coral Conservatory), and Australia (Great Barrier Reef Legacy). Due to the recognized small population status of *A. palmata* in Florida by 2022, NOAA solicited formal management recommendations for enhancing the redundancy of genets among ocean-based and land-based propagation facilities and genetic banks (Rodriguez-Clark et al., 2023). The same was done for *A. cervicornis*, although less formally, prompting the previously discussed coral genet swaps over the last decade. These actions, many taken prior to when general reef management paradigms might have deemed them needed, were key to enabling the preservation of this species. However, by 2023 most unique genets were not yet present in a dedicated coral biobank.

The emergency coral response and rescue resulted in the swift acquisition of as many remaining unique genets as possible from multiple restoration programs to populate 2 redundant

land-based coral biobanks: Mote's International Coral Gene Bank and The Reef Institute (Figures 1c & 2; Appendix S1). Similar efforts occurred with collections from the few remaining wild acroporids that were not present in any restoration nursery. These initial efforts alone filled quarantine spaces in the gene banks quickly, thus preventing the acquisition of more corals for safeguarding during this response period. The coordination of these multiple efforts, led by the NOAA recovery coordinator, within a short period was unprecedented. Having federal leaders coordinate multiple transfers among organizations to ensure that genets were banked prior to further loss took significant time, resources, and trust among state and federal management agencies and the restoration practitioners. The time and effort that these individuals put into building these relationships over the past 2 decades became a critical asset to the successful execution of this effort.

Acroporid biomass evacuation

Most land-based facilities with temperature-controlled systems were already serving as active coral propagation nurseries, making space limited, and alternative strategies were necessary to house and safeguard evacuated corals. To provide further support, several facilities that were not previously holding corals for restoration retrofitted their infrastructure to hold evacuated corals as part of this emergency response; thus, they became impromptu land-based nurseries or biorepositories that can be used in the future (Figure 1d). During this massive evacuation effort, some complications arose during and following the activities that caused substantial loss of coral (Appendix S2), likely a result of moving thermally stressed corals, stressful transport conditions (including densely packed biomass), and subsequent holding in land-based facilities at high densities.

Caring for corals in land-based facilities is extremely challenging, even for some of the more easily maintained species. Caribbean *Acropora* are known to be particularly challenging to care for in aquaria when completely healthy and require specific parameters, which were difficult to maintain for the thousands of fragments in emergency holding facilities with limited space, staff, and resources. Some restoration practitioners were experts in the care of corals in an ocean-based nursery environment but had limited experience working with corals in land-based tanks. Therefore, FWC and NOAA developed a novel relationship in which specialized coral aquarists (the Heat Response Team) from Association of Zoos and Aquariums (AZA)-accredited facilities across the United States (HeaRT, <https://wildlifeflorida.org/heat-team-responses/>) were deployed to the Florida Keys to provide expert evaluations and recommendations on the critical care of evacuated corals in land-based holding. Team members spent 2 weeks working closely with approved coral health veterinarians and mentored those restoration practitioners in need of training, providing a vital source of information during this critical time while helping to care for evacuated corals.

Reducing sunlight to lessen bleaching stress

Bleaching impacts can be magnified by high light levels because light and temperature act synergistically in causing bleaching (Fitt & Warner, 1995; Jokiel & Coles, 1990). Shading corals from direct, intense solar irradiance during a heat-stress event can slow coral bleaching (Tagliafico et al., 2022) and thus theoretically promote coral survival and health. During the emergency response, restoration practitioners and researchers piloted different approaches to reduce light to offset bleaching severity in ocean-based nurseries and reef sites. One routine practice implemented in the summer of 2023 was lowering the coral trees closer to the substrate, thus increasing the depth of the corals by 1–2 m. This provided protection for corals in the lower portion of the tree (~9.1 m), particularly in reefs in the Upper Florida Keys and South Florida, but often corals on the upper half of the tree still bleached (~6.1 m) (Figure 4a).

A new approach included the establishment of a deepwater nursery where coral trees from shallow nurseries (~6.1 to ~9.1 m) were relocated to deeper waters, approximately 21 m deep, to mitigate light impacts (Figure 4b). Other piloted strategies to reduce light included shading in nurseries using mesh fabrics (Figure 4c) or acrylic sheets (Figure 4d) fixed just above corals in reef sites. In some places, the mitigation of light was either too late or the heat stress too intense, resulting in acroporid mortality despite the effort. In other situations, it may have helped. For example, moving the shallow water nursery to deep water showed promise, with many acroporid corals recovering and surviving, but problems with too little light became evident within several months, highlighting the limitations of this solution. Deploying shades or reducing light by other means before the visible onset of symbiont loss could offer the best chance of success. In the absence of controlled experiments, it is difficult to confirm which methods were useful and what level of light reduction is needed to mitigate bleaching in a meaningful way in situ, especially with the level of temperature stress observed in the summer of 2023 for Florida's Coral Reef, but further interventions could be trialed.

Impact of the bleaching event on ocean-based nurseries and outplants

Mortality of corals that remained in ocean-based nurseries varied by location and in congruence with local heat exposures; nurseries in South Florida and the Upper Florida Keys displayed substantially less mortality (0–10% acroporid loss in South Florida to ~0–40% in the Upper Keys) compared with nurseries in the Middle and Lower Florida Keys (up to 100% acroporid loss) (Figure 5a,b; Appendix S3).

Similar to the corals that remained in the ocean-based nurseries, acroporid corals that were outplanted and the few remnant wild colonies in Florida's Coral Reef exhibited high levels of bleaching (Figure 5c,d) and subsequent mortality (Williams

et al., 2024). However, one site at Elbow Reef had multiple survivors, and additional acroporid survivors were observed in other sites throughout the Florida Keys. Regional differences were also apparent, with little to no mortality in places, such as Miami-Dade (north of Fowey Rocks) and Broward Counties. Thus, outplanting across multiple environmental gradients (i.e., reef sites) can minimize losses. Although detailed data describing the levels of loss are described elsewhere (Williams et al., 2024), 100% loss of acroporids from previous individual outplant events was often observed. Although much of the outplanted coral perished throughout the Florida Keys, these outplanted sites were a hub for assisted sexual reproduction efforts while the corals were still alive (Williams et al., 2023), which included the summer of 2023, and these corals provided important ecosystem services while they occupied the reef.

Cost considerations

The response efforts undertaken by Florida's restoration network amounted to an estimated total of \$US2 million, primarily associated with moving corals among nurseries, land-based facility infrastructure, deeper ocean-based nurseries, 6 months of coral husbandry at land-based facilities, and veterinary health inspections upon the coral's return to the ocean. These estimated costs were in addition to the tens of millions of dollars allocated to restoration initiatives such as "Mission: Iconic Reefs" and other long-standing investments by members of the restoration network. As the coral evacuation process was considered extremely costly and resulted in low survival in the regions affected the most by the bleaching event (Appendix S2), a proactive rather than reactive approach is recommended.

The costly nature of coral reef restoration, the funding directed toward responding to the 2023 heatwave, and the detrimental effects of this and similar disturbances raise the question of whether the benefits of restoration outweigh the costs (Hughes et al., 2023). The record-breaking 2023 thermal anomaly demonstrated that without prior restoration efforts, which included sustained investment in both ocean-based and land-based infrastructure coupled with rapid and effective action, *A. palmata* and *A. cervicornis* would have been functionally extirpated, meaning there would have been no remaining stocks to preserve the species regionally or repopulate local reefs. [Correction added on December 2, 2025 after first online publication: The section "Cost considerations" has been moved here from its previous location.]

LESSONS LEARNED AND NEXT STEPS

The primary lesson among many learned from the restoration community's response to the 2023 heatwave was that the previous 20 years of effort prevented the regional loss of 2 threatened functionally important coral species, *A. palmata* and *A. cervicornis*, and preserved critical genetic richness to ensure the possibility of regional species recovery. Given what was learned and

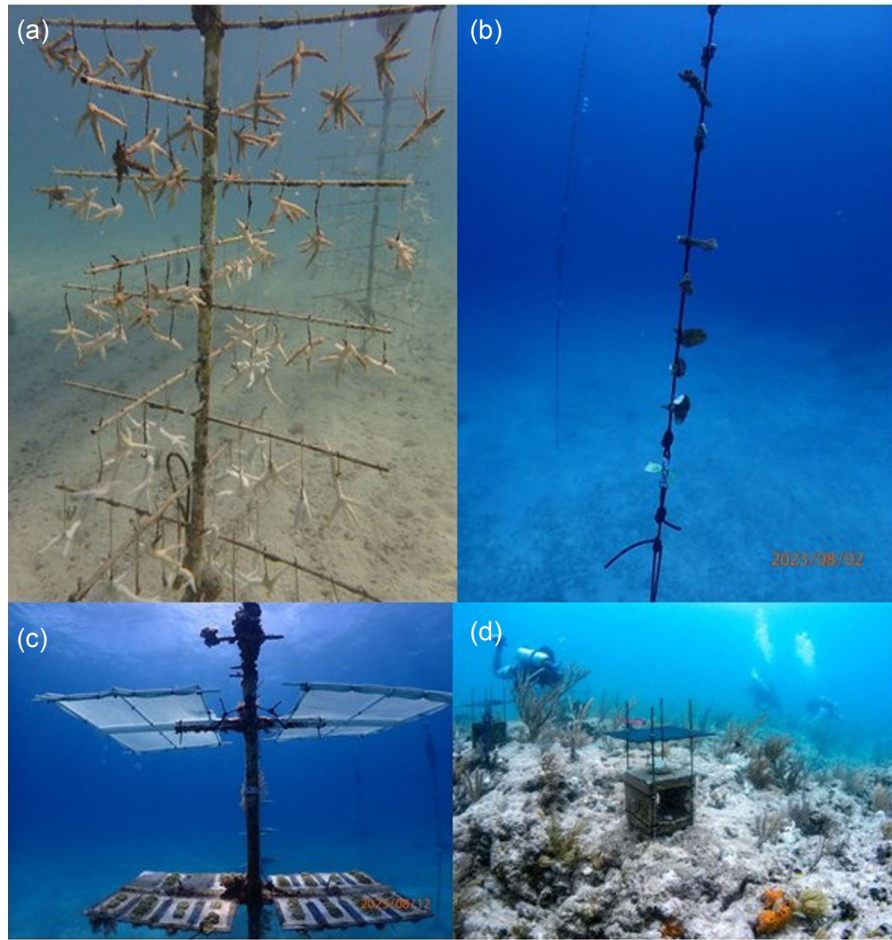


FIGURE 4 Examples of shading structures or techniques piloted to reduce bleaching impacts on *Acropora* spp. during the summer of 2023: (a) coral tree near the substrate, with dead coral fragments being colonized by algae on the top of the tree and living but completely bleached corals on the bottom of the tree (photo by Mote Marine Laboratory), (b) deep water nurseries (photo by Reef Renewal), (c) shaded panels on underwater tree structures (photo by Reef Renewal), and (d) acrylic shade panels placed over *A. palmata* corals during an experiment (photo by USGS).

the expanding geographical scope of extreme heat stress, we suggest other regions, particularly in the Caribbean, could consider preparing for similar disasters by developing a network of trained experts; establishing trusted relationships between management agencies and restoration groups; trialing direct interventions to reduce light/temperature stress during thermal anomalies early; developing redundant ocean-based and land-based nurseries; and establishing living coral biobanks prior to major threats to prevent the extirpation of coral species. Additional interventions that have not been trialed in Florida could also be considered because there is likely no universal best option due to differing priorities, opportunities, and limitations in reef regions around the world.

Building a network of trained experts

A previously established network of trained coral aquarists from restoration practitioners and AZA-accredited facilities resulting from extensive investment and activity in coral culture and restoration over the last 20 years proved advantageous

during the coral evacuation and care process. However, establishing a broader international network of experts throughout a species' range would be most effective. In Florida, where arguably the most active coral restoration is occurring in the United States, there are still not enough staff well trained in land-based coral care and too few coral health veterinarians to meet the current production needs. As such, there are new hands-on training courses being developed by AZA and restoration partners to expand the skilled workforce and facilitate the transfer of knowledge and expertise critical to acroporid species preservation range wide.

Identifying champions and establishing trusted relationships

As part of NOAA's mandate to recover ESA-listed species, a recovery coordinator has been facilitating collaborative actions for the conservation of the acroporid species for over 21 years. Having one champion who was aware of the breadth of conservation organizations and actions fostered collaboration,

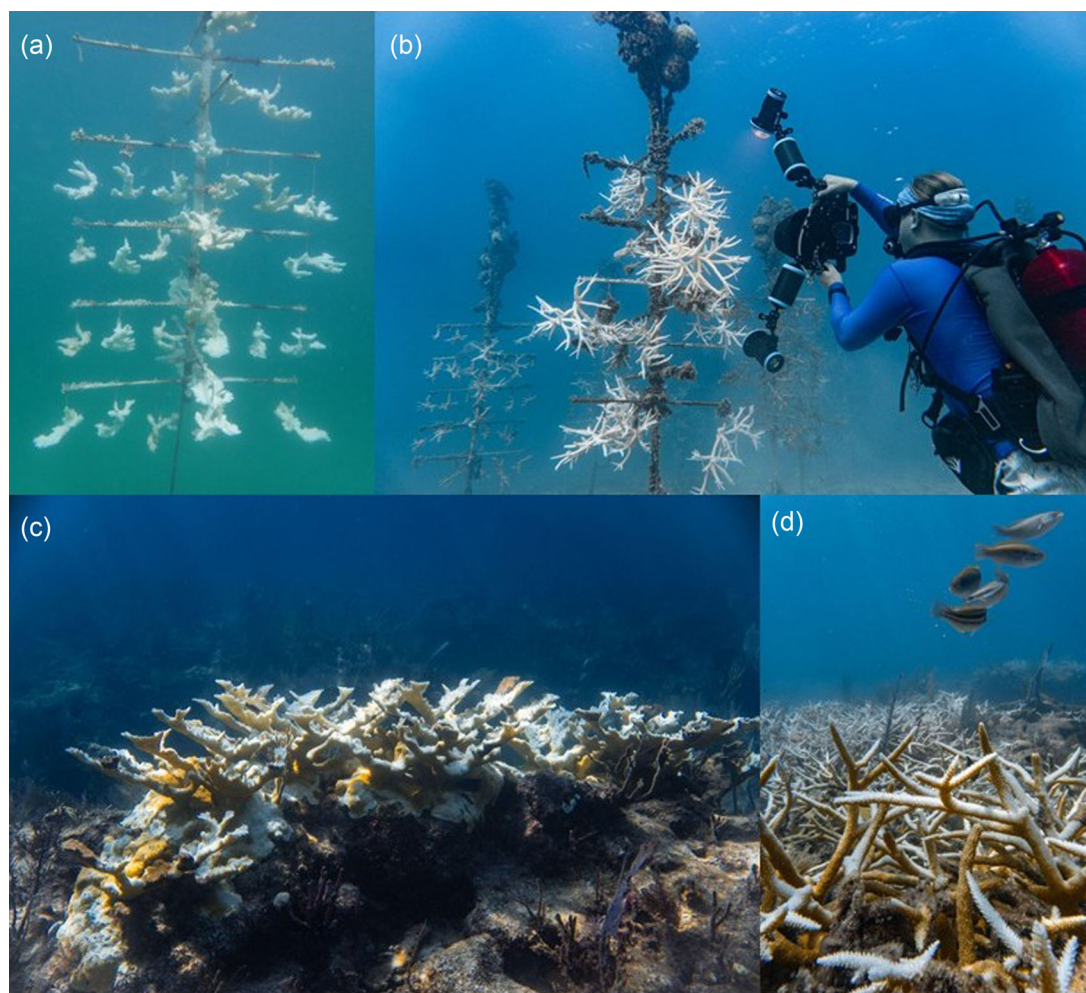


FIGURE 5 Bleached acroporid corals in underwater nurseries: (a) *Acropora palmata* (photo by Mote Marine Laboratory), (b) *Acropora cervicornis* at outplant sites, (c) *Acropora palmata*, and (d) *Acropora cervicornis*. Photos in (b), (c), and (d) by Coral Restoration Foundation.

efficiency, and efficacy to make the efforts of each more powerful. Similarly, management agencies and restoration groups need a mutual level of trust and respect to coordinate efforts and allow for adaptive, response-based management. Although regulatory oversight increased during the coral evacuation and return process and continues to increase, the emergency response would not have happened without the coordination and mutual trust among groups. For example, these relationships facilitated the establishment of novel deepwater nurseries that required emergency authorization from multiple federal agencies (e.g., U.S. Coast Guard, NOAA). This process took days rather than months, enabling a mitigation strategy that was arguably more effective and less costly than moving large amounts of acroporids into land-based systems.

Establishing interventions early

Prior to high summer temperatures, it would be helpful to establish options with management agency partners for tri-

aling methods to reduce light and temperature stress during periods of warming. Options could include deep water nurseries, direct shading, microbubbles, artificial cooling, marine cloud brightening, and other novel techniques that are currently under investigation. These options are now being considered for ocean-based coral nurseries as well as high-value outplant sites.

Importance of redundancy

A network of established nurseries throughout a large geographic range of the reef tract provided several opportunities for acroporid corals to escape the thermal stress, which was more extreme in the Lower Florida Keys compared with the Upper Keys and Southeast Florida. Such redundancy would be equally beneficial for other threats such as disease outbreaks or hurricanes. This recommendation is also the easiest to implement in regions with limited access to land-based infrastructure, reliable power, or boats. Having multiple ocean-based and land-based nurseries with as many genets as possible spread throughout dif-

ferent environmental regions spreads the risk associated with losing any genet and will potentially retain biomass. Disease outbreaks can affect land-based systems as well, especially in aquaculture facilities with high stocking densities. Therefore, ensuring redundancy spatially and in a diversity of holding types will provide the best fail-safe protection against known and unknown threats.

Creating redundant safe havens

The living coral biobanks were essential to the preservation of a substantial subset of genets in the threatened populations prior to the 2023 event. Taking preemptive, methodically planned actions rather than a reactionary approach reduced logistical hurdles, allowed for better preparation and efficacy, facilitated coordination, provided a greater chance of coral survival during the transfer and acclimation period, and reduced the likelihood of costly mistakes.

LOOKING AHEAD

The foundation laid by coral restoration practitioners has facilitated essential research on actionable interventions that can be employed to enhance resilience to threats, particularly due to climate stressors. Given the marine heatwave of 2023, effective and timely translation of that research is critical for acroporids to survive even present-day conditions in Florida. Existing research and observations show promise that interventions, such as selective breeding (Drury et al., 2022), algal symbiont manipulation (Cunning et al., 2018; Elder et al., 2023; Silverstein et al., 2015), microbiome manipulations (De Breuyn et al., 2025), conservation translocations (Kuffner et al., 2020), and assisted gene flow (Hagedorn et al., 2021), could increase resilience to thermal stress. Continued research on the genetic basis and heritability of traits, such as heat tolerance, disease resistance, and immune capabilities, is essential to further guide population management plans and restoration strategies (Kenkel et al., 2013; Muller et al., 2018). However, testing these interventions in the wild and scaling up are essential to finding solutions that can be implemented at meaningful scales and in time to keep pace with changing ocean conditions. In the meantime, interventions that are necessary to preserve existing genetic lines, including living coral biobanks (utilizing both aquaculture and cryopreservation techniques), land-based and ocean-based nurseries, and assisted sexual reproduction, are essential for the persistence of these threatened species as ocean conditions continue to change (Rodriguez-Clark et al., 2023). Although the efforts described above have bought corals time and, so far, preserved regionally important populations of 2 critical species, there is a need to further advance interventions to promote the recovery of these species on Florida's Coral Reef in the near term. Simultaneously, the persistence and functioning of coral reef ecosystems, regionally and globally, may depend on the trajectory of human action on the drivers of climate change (Reimer et al., 2024).

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DATA AVAILABILITY STATEMENT

All data used to draw conclusions in this paper are in the Supporting Information Appendices.

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REFERENCES

- Boisvert, T. S., Ruzicka, R. R., Schopmeyer, S. A., & Stallings, C. D. (2024). Restoration success limited by poor long-term survival after 9 years of *Acropora cervicornis* outplanting in the upper Florida Keys, United States. *Restoration Ecology*, 32(4), Article e14129. <https://doi.org/10.1111/rec.14129>
- Boström-Einarsson, L., Babcock, R. C., Bayraktarov, E., Ceccarelli, D., Cook, N., Ferse, S. C. A., Hancock, B., Harrison, P., Hein, M., Shaver, E., Smith, A., Suggett, D., Stewart-Sinclair, P. J., Vardi, T., & McLeod, I. M. (2020). Coral restoration—A systematic review of current methods, successes, failures and future directions. *PLoS ONE*, 15(1), Article e0226631.
- Bruckner, A. W. (2002). *Proceedings of the Caribbean Acropora Workshop: Potential application of the USA Endangered Species Act as a conservation strategy* (Technical memorandum NMFS-OPR-24). NOAA.
- Carlson, R. R., Foo, S. A., & Asner, G. P. (2019). Land use impacts on coral reef health: A ridge-to-reef perspective. *Frontiers in Marine Science*, 6, Article 562. <https://doi.org/10.3389/fmars.2019.00562>
- Carpenter, K. E., Abrar, M., Aeby, G., Aronson, R. B., Banks, S., Bruckner, A., Chiriboga, A., Cortés, J., Delbeek, J. C., Devantier, L., Edgar, G. J., Edwards, A. J., Fenner, D., Guzmán, H. M., Hoeksema, B. W., Hodgson, G., Johan, O., Licuanan, W. Y., Livingstone, S. R., ... Wood, E. (2008). One-third of reef-building corals face elevated extinction risk from climate change and local impacts. *Science*, 321(5888), 560–563.
- Carrick, J. V., Lusic, C., Lirman, D. V., Schopmeyer, S., Bartels, E., Burdeno, D., Dahlgren, C., Galvan, V., Gilliam, D. S., Goergen, E., Gore, S., & Griffin, S. (2019, November 3–7). *Hurricane impacts on reef restoration: The good, the bad, and the ugly* [Conference presentation]. 2019 CERF Biennial Conference, Mobile, AL, United States. <https://cerf.confex.com/cerf/2019/meetingapp.cgi/Paper/4992>

- Colella, M. A., Ruzicka, R. R., Kidney, J. A., Morrison, J. M., & Brinkhuis, V. B. (2012). Cold-water event of January 2010 results in catastrophic benthic mortality on patch reefs in the Florida Keys. *Coral Reefs*, 31(2), 621–632.
- Cook, N., Smith, A., Songcuan, A., Cassidy, D., Sartori, G., & McLeod, I. (2022). Lessons learned from small-scale coral outplanting intervention at a restoration site on the Great Barrier Reef. *Ecological Management & Restoration*, 23(1), 89–93.
- Craggs, J., Guest, J., Davis, M., & Sweet, M. (2020). Completing the life cycle of a broadcast spawning coral in a closed mesocosm. *Invertebrate Reproduction & Development*, 64(3), 244–247.
- Cunning, R., Silverstein, R. N., & Baker, A. C. (2018). Symbiont shuffling linked to differential photochemical dynamics of Symbiodinium in three Caribbean reef corals. *Coral Reefs*, 37(1), 145–152.
- De Breuyn, M., Ostendarp, M., El-Khaled, Y. C., Garcias-Bonet, N., Carvalho, S., Wild, C., & Peixoto, R. S. (2025). Probiotics prevent mortality of thermal-sensitive corals exposed to short-term heat stress. *ISME Communications*, 5(1), Article ycaf039.
- Drury, C., Caruso, C., & Quigley, K. (2022). Selective breeding to enhance the adaptive potential of corals. In M. J. H. van Oppen & M. A. Lastra (Eds.), *Coral reef conservation and restoration in the omics age* (pp. 71–84). Springer.
- Elder, H., Million, W. C., Bartels, E., Krediet, C. J., Muller, E. M., & Kenkel, C. D. (2023). Long-term maintenance of a heterologous symbiont association in *Acropora palmata* on natural reefs. *The ISME Journal*, 17(3), 486–489.
- Fitt, W. K., & Warner, M. E. (1995). Bleaching patterns of four species of Caribbean reef corals. *The Biological Bulletin*, 189(3), 298–307.
- Hagedorn, M., Page, C. A., O'Neil, K. L., Flores, D. M., Tichy, L., Conn, T., Chamberland, V. F., Lager, C., Zuchowicz, N., Lohr, K., Blackburn, H., Vardi, T., Moore, J., Moore, T., Baums, I. B., Vermeij, M. J. A., & Marhaver, K. L. (2021). Assisted gene flow using cryopreserved sperm in critically endangered coral. *Proceedings of the National Academy of Sciences of the United States of America*, 118(38), Article e2110559118. <https://doi.org/10.1073/pnas.2110559118>
- Harper, L. M., Huebner, L. K., O'Cain, E. D., Ruzicka, R., Gleason, D. F., & Fogarty, N. D. (2023). Multi-year coral recruitment study across the Florida Reef Tract reveals boom-or-bust pattern among broadcast spawners and consistency among brooders. *Marine Ecology Progress Series*, 721, 39–58.
- Hogarth, W. T. (2006). Endangered and threatened species: Final listing determinations for elkhorn coral and staghorn coral. *Federal Register*, 71, 26852–26861.
- Hughes, T. P., Baird, A. H., Morrison, T. H., & Torda, G. (2023). Principles for coral reef restoration in the Anthropocene. *One Earth*, 6(6), 656–665.
- Johnson, M. E., Lusic, C., Bartels, E., Baums, I. B., Gilliam, D. S., Larson, E. A., Lirman, D., Miller, M. W., Nedimyer, K., & Schopmeyer, S. (2011). *Caribbean Acropora restoration guide: Best practices for propagation and population enhancement*. The Nature Conservancy.
- Jokiel, P. L., & Coles, S. L. (1990). Response of Hawaiian and other Indo-Pacific reef corals to elevated temperature. *Coral Reefs*, 8(4), 155–162.
- Jones, N. P., Ruzicka, R. R., Colella, M. A., Pratchett, M. S., & Gilliam, D. S. (2022). Frequent disturbances and chronic pressures constrain stony coral recovery on Florida's Coral Reef. *Coral Reefs*, 41(6), 1665–1679.
- Kelly, E., Gidley, M., Sinigalliano, C., Kumar, N., & Solo-Gabriele, H. M. (2021). Impact of wastewater infrastructure improvements on beach water fecal indicator bacteria levels in Monroe County, Florida. *Science of the Total Environment*, 763, Article 143024.
- Kenkel, C. D., Goodbody-Gringley, G., Caillaud, D., Davies, S. W., Bartels, E., & Matz, M. V. (2013). Evidence for a host role in thermotolerance divergence between populations of the mustard hill coral (*Porites astreoides*) from different reef environments. *Molecular Ecology*, 22(16), 4335–4348.
- Kitchen, S. A., Von Kuster, G., Kuntz, K. V., Reich, H. G., Miller, W., Griffin, S., Fogarty, N. D., & Baums, I. B. (2020). STAGdb: A 30K SNP genotyping array and Science Gateway for *Acropora* corals and their dinoflagellate symbionts. *Scientific Reports*, 10(1), Article 12488.
- Kleypas, J., Allemand, D., Anthony, K., Baker, A. C., Beck, M. W., Hale, L. Z., Hilmi, N., Hoegh-Guldberg, O., Hughes, T., Kaufman, L., Kayanne, H., Magnan, A. K., McLeod, E., Mumby, P., Palumbi, S., Richmond, R. H., Rinkevich, B., Steneck, R. S., Voolstra, C. R., ... Gattuso, J.-P. (2021). Designing a blueprint for coral reef survival. *Biological Conservation*, 257, Article 109107.
- Kuffner, I. B., Stathakopoulos, A., Toth, L. T., & Bartlett, L. A. (2020). Reestablishing a stepping-stone population of the threatened elkhorn coral *Acropora palmata* to aid regional recovery. *Endangered Species Research*, 43, 461–473.
- Lirman, D., & Schopmeyer, S. (2016). Ecological solutions to reef degradation: Optimizing coral reef restoration in the Caribbean and Western Atlantic. *PeerJ*, 4, Article e2597.
- Lirman, D., Schopmeyer, S., Manzello, D., Gramer, L. J., Precht, W. F., Muller-Karger, F., Banks, K., Barnes, B., Bartels, E., Bourque, A., Byrne, J., Donahue, S., Duquesnel, J., Fisher, L., Gilliam, D., Hendee, J., Johnson, M., Maxwell, K., McDevitt, E., ... Thanner, S. (2011). Severe 2010 cold-water event caused unprecedented mortality to corals of the Florida reef tract and reversed previous survivorship patterns. *PLoS ONE*, 6(8), Article e23047.
- Lirman, D., Thyberg, T., Herlan, J., Hill, C., Young-Lahiff, C., Schopmeyer, S., Huntington, B., Santos, R., & Drury, C. (2010). Propagation of the threatened staghorn coral *Acropora cervicornis*: Methods to minimize the impacts of fragment collection and maximize production. *Coral Reefs*, 29(3), 729–735.
- Manzello, D. P. (2015). Rapid recent warming of coral reefs in the Florida Keys. *Scientific Reports*, 5, Article 16762.
- Manzello, D. P., Berkemans, R., & Hendee, J. C. (2007). Coral bleaching indices and thresholds for the Florida Reef Tract, Bahamas, and St. Croix, US Virgin Islands. *Marine Pollution Bulletin*, 54(12), 1923–1931.
- McLeod, I. M., Hein, M. Y., Babcock, R., Bay, L., Bourne, D. G., Cook, N., Doropoulos, C., Gibbs, M., Harrison, P., Lockie, S., van Oppen, M. J. H., Mattocks, N., Page, C. A., Randall, C. J., Smith, A., Smith, H. A., Suggett, D. J., Taylor, B., Vella, K. J., ... Boström-Einarsson, L. (2022). Coral restoration and adaptation in Australia: The first five years. *PLoS ONE*, 17(11), Article e0273325.
- Miller, M. W., Kerr, K., & Williams, D. E. (2016). Reef-scale trends in Florida *Acropora* spp. abundance and the effects of population enhancement. *PeerJ*, 4, Article e2523.
- Muehlhner, N., Langdon, C., Venti, A., & Kadko, D. (2016). Dynamics of carbonate chemistry, production, and calcification of the Florida Reef Tract (2009–2010): Evidence for seasonal dissolution. *Global Biogeochemical Cycles*, 30(5), 661–688.
- Muller, E. M., Bartels, E., & Baums, I. B. (2018). Bleaching causes loss of disease resistance within the threatened coral species *Acropora cervicornis*. *eLife*, 7, Article e35066. <https://doi.org/10.7554/eLife.35066>
- Muller, E. M., Sartor, C., Alcaraz, N. I., & van Woesik, R. (2020). Spatial epidemiology of the stony-coral-tissue-loss disease in Florida. *Frontiers in Marine Science*, 7, Article 163. <https://doi.org/10.3389/fmars.2020.00163>
- Nalley, E. M., Tuttle, L. J., Barkman, A. L., Conklin, E. E., Wulstein, D. M., Richmond, R. H., & Donahue, M. J. (2021). Water quality thresholds for coastal contaminant impacts on corals: A systematic review and meta-analysis. *The Science of the Total Environment*, 794, Article 148632.
- National Oceanic and Atmospheric Administration (NOAA). (2015). *Recovery plan for elkhorn (Acropora palmata) and staghorn (A. cervicornis) corals*. National Marine Fisheries Service. <https://repository.library.noaa.gov/view/noaa/8950>
- Nedimyer, K., Gaines, K., & Roach, S. (2011). Coral Tree Nursery: An innovative approach to growing corals in an ocean-based field nursery. *Aquaculture, Aquarium, Conservation & Legislation*, 4(4), 442–446.
- Porter, J., & Meier, O. W. (1992). Quantification of loss and change in Floridian reef coral populations. *Integrative and Comparative Biology*, 32, 625–640.
- Porter, J. W., Dustan, P., Jaap, W. C., Patterson, K. L., Kosmynin, V., Meier, O. W., Patterson, M. E., & Parsons, M. (2001). Patterns of spread of coral disease in the Florida Keys. In J. W. Porter (Ed.), *The ecology and etiology of newly emerging marine diseases* (pp. 1–24). Springer.
- Precht, W. F., Aronson, R. B., Miller, S. L., Keller, B. D., & Causey, B. (2005). The folly of coral restoration programs following natural disturbances in the Florida Keys national marine sanctuary. *Ecological Restoration*, 23(1), 24–28.
- Precht, W. F., Gintert, B. E., Robbatt, M. L., Fura, R., & van Woesik, R. (2016). Unprecedented disease-related coral mortality in southeastern Florida. *Scientific Reports*, 6, Article 31374.

- Reimer, J. D., Peixoto, R. S., Davies, S. W., Traylor-Knowles, N., Short, M. L., Cabral-Tena, R. A., Burt, J. A., Pessoa, I., Banaszak, A. T., Winters, R. S., Moore, T., Schoepf, V., Kaulysing, D., Calderon-Aguilera, L. E., Wörheide, G., Harding, S., Munbodhe, V., Mayfield, A., Ainsworth, T., ... Voolstra, C. R. (2024). The Fourth Global Coral Bleaching Event: Where do we go from here *Coral Reefs*, 43, 1121–1125. <https://doi.org/10.1007/s00338-024-02504-w>
- Rodriguez-Clark, K. M., Moore, J., Baums, I., Ferrie, G., Hagedorn, M., Miller, M., & Willis, K. (2023). *Population Management of Elkhorn Coral (Acropora palmata) for F3P: Part II, Recommendations and Guidelines Jun 2023-Feb 2024*. Smithsonian's National Zoo and Conservation Biology Institute.
- Ruzicka, R. R., Colella, M. A., Porter, J. W., Morrison, J. M., Kidney, J. A., Brinkhuis, V., Lunz, K. S., Macaulay, K. A., Bartlett, L. A., Meyers, M. K., & Colee, J. (2013). Temporal changes in benthic assemblages on Florida Keys reefs 11 years after the 1997/1998 El Niño. *Marine Ecology Progress Series*, 489, 125–141.
- Schopmeyer, S. A., Lirman, D., & Bartels, E. (2012). In situ coral nurseries serve as genetic repositories for coral reef restoration after an extreme cold-water event. *Restoration*, 20(6), 696–703. <https://doi.org/10.1111/j.1526-100X.2011.00836.x>
- Sheridan, C., Kramarsky-Winter, E., Sweet, M., Kushmaro, A., & Leal, M. C. (2013). Diseases in coral aquaculture: Causes, implications and preventions. *Aquaculture*, 396–399, 124–135.
- Silverstein, R. N., Cunnning, R., & Baker, A. C. (2015). Change in algal symbiont communities after bleaching, not prior heat exposure, increases heat tolerance of reef corals. *Global Change Biology*, 21(1), 236–249.
- Somerfield, P. J., Jaap, W. C., Clarke, K. R., Callahan, M., Hackett, K., Porter, J., Lybolt, M., Tsokos, C., & Yanev, G. (2008). Changes in coral reef communities among the Florida Keys, 1996–2003. *Coral Reefs*, 27(4), 951–965.
- Swaminathan, S. D., Lafferty, K. D., Knight, N. S., & Altieri, A. H. (2024). Stony coral tissue loss disease indirectly alters reef communities. *Science Advances*, 10(18), Article eadk6808.
- Tagliafico, A., Baker, P., Kelaher, B., Ellis, S., & Harrison, D. (2022). The effects of shade and light on corals in the context of coral bleaching and shading technologies. *Frontiers in Marine Science*, 9, Article 919382. <https://doi.org/10.3389/fmars.2022.919382>
- Toth, L. T., Storlazzi, C. D., Kuffner, I. B., Quataert, E., Reyns, J., McCall, R., Stathakopoulos, A., Hillis-Starr, Z., Holloway, N. H., Ewen, K. A., Pollock, C. G., Code, T., & Aronson, R. B. (2023). The potential for coral reef restoration to mitigate coastal flooding as sea levels rise. *Nature Communications*, 14(1), Article 2313.
- Toth, L. T., van Woesik, R., Murdoch, T. J. T., Smith, S. R., Ogden, J. C., Precht, W. F., & Aronson, R. B. (2014). Do no-take reserves benefit Florida's corals? 14 years of change and stasis in the Florida Keys National Marine Sanctuary. *Coral Reefs*, 33(3), 565–577.
- van Woesik, R., Banister, R. B., Bartels, E., Gilliam, D. S., Goergen, E. A., Lustic, C., Maxwell, K., Moura, A., Muller, E. M., Schopmeyer, S., Winters, R. S., & Lirman, D. (2021). Differential survival of nursery-reared *Acropora cervicornis* outplants along the Florida reef tract. *Restoration Ecology*, 29(1), Article e13302. <https://doi.org/10.1111/rec.13302>
- Wagner, D. E., Kramer, P., & van Woesik, R. (2010). Species composition, habitat, and water quality influence coral bleaching in southern Florida. *Marine Ecology Progress Series*, 408, 65–78.
- Ware, M., Garfield, E. N., Nedimyer, K., Levy, J., Kaufman, L., Precht, W., Winters, R. S., & Miller, S. L. (2020). Survivorship and growth in staghorn coral (*Acropora cervicornis*) outplanting projects in the Florida Keys National Marine Sanctuary. *PLoS ONE*, 15(5), Article e0231817.
- Williams, D. E., Bright, A. J., Williamson, O. M., Moura, A., & Ladd, M. C. (2023). Synchronous spawning of nursery-raised elkhorn coral (*Acropora palmata*) outplanted to reefs in the Florida Keys (United States). *Frontiers in Marine Science*, 10, Article 1180996. <https://doi.org/10.3389/fmars.2023.1180996>
- Williams, D. E., Miller, M. W., Bright, A. J., Pausch, R. E., & Valdivia, A. (2017). Thermal stress exposure, bleaching response, and mortality in the threatened coral *Acropora palmata*. *Marine Pollution Bulletin*, 124(1), 189–197.
- Williams, D. E., Miller, M. W., & Kramer, K. L. (2008). Recruitment failure in Florida Keys *Acropora palmata*, a threatened Caribbean coral. *Coral Reefs*, 27(3), 697–705.
- Williams, D. E., Nedimyer, K., Bright, A. J., & Ladd, M. C. (2024). *Genotypic inventory and impact of the 2023 marine heatwave on Acropora palmata (elkhorn coral) populations in the Upper Florida Keys, USA: 2020–2023*. NOAA Fisheries SEFSC. <https://doi.org/10.25923/37c0-x182>
- Williams, D. E., Nedimyer, K., & Miller, M. (2020). *Genotypic inventory of Acropora palmata (elkhorn coral) populations in South Florida* (NOAA/SEFSC/PRBD-2020-1). NOAA National Marine Fisheries Service.
- Zaneveld, J. R., Burkepille, D. E., Shantz, A. A., Pritchard, C. E., McMinds, R., Payet, J. P., Welsh, R., Correa, A. M. S., Lemoine, N. P., Rosales, S., Fuchs, C., Maynard, J. A., & Thurber, R. V. (2016). Overfishing and nutrient pollution interact with temperature to disrupt coral reefs down to microbial scales. *Nature Communications*, 7, Article 11833.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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